Background-Guided Deformable Convolutional Autoencoder for Hyperspectral Anomaly Detection

Zhaoyue Wu, Mercedes E. Paoletti, *Senior Member, IEEE*, Hongjun Su, *Senior Member, IEEE*, Xuanwen Tao, *Member, IEEE*, Lirong Han, Juan M. Haut, *Senior Member, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— Autoencoder (AE)-based hyperspectral anomaly detectors have received significant attention. The core of these detectors is to reconstruct backgrounds by optimizing AEs so that anomalies can be detected by reconstruction residuals. Nevertheless, existing methods are flawed in two aspects: 1) most of them reconstruct the background along with the anomalies, resulting in undesired performance for large target detection in complex backgrounds, and 2) they only focus on the encoder optimization part, ignoring the decoder reconstruction quality of the background. Given the above, this article proposes a background-guided deformable convolutional AE (`DCAE`) network with three mutually supportive parts, including encoder, decoder, and background guidance modules. In the encoder, deformable convolution (DC) is introduced into regular convolution to build the adaptive spatial feature extractor to fit complex spatial structures, while a nonlocal convolution (NLC) is introduced to build an external feature extractor to capture global spatial relationships. Furthermore, a mask is designed to filter potential anomalous information, curbing the representation of high-frequency anomalies to focus on widespread backgrounds. In the decoder, a background guidance module (considering the physical meaning of linear reconstruction) is built, guiding the proposed network learning via two strategies. One is initializing the weight of the decoder, and another is adding a loss term. Notably, both the number of output channels of the encoder and the decoder construction are determined by the background guidance module, which creates a bridge between the network design and practical situations. A profound analysis demonstrates the outstanding performance of the proposed method, which outperforms traditional and deep learning methods, proving that the novel designs introduced in the network architecture are extremely effective.**

***Index Terms*— Autoencoder (AE), background guidance, deformable convolution (DC), hyperspectral anomaly detection, nonlocal convolution (NLC).**

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Zhaoyue Wu, Mercedes E. Paoletti, Xuanwen Tao, Lirong Han, Juan M. Haut, and Antonio Plaza are with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, 10003 Cáceres, Spain (e-mail: zhaoyue_wu@163.com; mpaoletti@unex.es; taoxuanwenupc@gmail.com; lironghan_upc@163.com; juanmariohaut@unex.es; aplaza@unex.es).

Hongjun Su is with the College of Geography and Remote Sensing, Hohai University, Nanjing 211100, China (e-mail: hjsu@hhu.edu.cn).

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I. Introduction

HYPERSPECTRAL imaging is a hybrid modality of imaging developed by combining imaging and spectroscopic sciences, which generates 3-D data (called hypercube) where each pixel collects fine spectral information [1], [2], [3]. Compared with other types of image data, hyperspectral images (HSIs) have higher spectral resolution along with detailed spatial information to reflect the chemical composition and physical pattern of the objects [4], [5], [6]. This results in unique advantages in many applications.

Hyperspectral anomaly detection aims to determine whether there is an anomalous target in the scene, based on the spectral difference between the target and the background [7]. Until now, the definition of anomalous targets remains ambiguous. These targets are generally considered to deviate from the normal background clutter distribution, having characteristics such as low probability of occurrence and small area coverage [8]. Depending on the application scenario, anomalous targets may appear as different entities, such as infected trees in woods, rare minerals in geological applications, and man-made objects (vehicles, aircraft, and tanks) in defense applications. Therefore, the core task of hyperspectral anomaly detection is to effectively separate the anomalous target (low probability of occurrence and small area) from the background (extensive and complex distribution), according to their features [9]. The existing hyperspectral anomaly detectors can be mainly categorized into traditional methods and deep-learning-based methods.

A. Traditional Anomaly Detectors

These methods mainly include statistics-based and representation-based approaches. The statistics-based HSI anomaly detection methods are designed according to the distribution differences between anomalies and backgrounds. Among them, the Reed–Xiaoli (RX) detector [10] is a well-known and representative statistical method. RX makes an assumption of the background obeying a multivariate normal distribution, where the probability of the pixel under test (PUT) being anomalous is estimated, i.e., the more the PUT deviates from a Gaussian distribution, the more likely it is to be an anomalous target. Extensive experiments have shown that RX performs satisfactorily in HSIs with simple backgrounds, but not in complex scenes. In view of this, many improved versions of the RX algorithm have been

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developed [11], [12], [13]. Recently, some simple and novel detectors—inspired by the idea of isolated forests (iForest)—have stepped into the limelight. They detect anomalous targets by constructing many isolated trees composed of randomly selected pixels [14], [15], [16], [17], which can yield excellent results on small target detection tasks.

Hyperspectral anomaly detection algorithms based on representation models consider that all the background pixels can be reconstructed by a well-designed background dictionary, while the anomalous pixels cannot. Depending on the processing unit, the representation models can be divided into pixel-by-pixel and whole-image representation ones. The former ones select the surrounding pixels as a background dictionary to reconstruct the PUT, such as sparse representation (SR) detectors [18], [19] and collaborative representation (CR) detectors [20], [21], [22], in which sliding dual windows are normally adopted to build the dictionary. This approach has been extensively studied and can achieve good results given a pure background dictionary. The latter assumes that the whole image can be separated into two parts: the background and the anomaly, where a global background dictionary is built to reconstruct all types of backgrounds. Furthermore, given the fact that the background is widely distributed while the target rarely appears, low-rank and sparse constraints are assigned to the background and the target, respectively. Some representative methods include low-rank and SR detector (LRASR) [23], low-rank matrix decomposition based on Mahalanobis distance for anomaly detector (LSMAD) [24], low-rank and CR detector (LRCRD) [25], and so on [26], [27], [28]. Moreover, to preserve the inherent spatial structure of the hyperspectral 3-D cube, extensive methods using tensor decomposition have been developed. Examples include structure tensor and guided-filter-based detector [29], prior-based tensor approximation detector (PTA) [30], and tensor low-rank and SR for hyperspectral anomaly detection [31], [32], [33], [34], among many others. Numerous studies have demonstrated that the low-rank constraint is essential for hyperspectral anomaly detection, allowing the model to effectively distinguish background and anomaly under a delicately designed dictionary.

B. Deep-Learning-Based Anomaly Detectors

In recent years, deep learning has shown tremendous capabilities in various applications such as language recognition, image processing, and video tracking, evolving lots of promising networks. In terms of hyperspectral anomaly detection, convolutional neural networks (CNNs), autoencoders (AEs), and generative adversarial networks (GANs) have been extensively studied. CNN-based anomaly detectors are usually designed using some available information, such as transferred CNN which detects anomalous targets by learning an image with classification labels acquired by the same sensor [35], [36]. The latest CNNs provide a new solution for hyperspectral anomaly detection which designs a blind spot strategy to train a background reconstruction network to detect anomalies directly [37], [38].

The AE-based anomaly detectors consider that anomalies are large residuals in the process of automatic image reconstruction due to their distinctiveness and rarity. It is worth noting that the behavioral patterns of the background dominating the image can be more easily learned (and reconstructed) by a well-designed AE network than that of anomalies. Therefore, the key goal is to design an AE network that almost exclusively reconstructs the background. Lei et al. [39] impose a constraint on the encoder to ensure that the encoded feature obeys the Gaussian distribution, purifying the background. Xie et al. [40] adopt the Markov chain Monte Carlo and Gaussian mixture model so that the network adapts to complex scenes. Li et al. [41] combine a variational AE and a normalizing flow to transform the Gaussian distribution into a more complex distribution, resulting in a more generalizable model. To learn intrinsic low-dimensional attributes of HSIs, Jiang et al. [42] introduce a local invariant AE to enhance the low-dimensional representation. Wang et al. [43] build a dense residual self-attention module and a low-rank loss function to constrain the generation of the latent features. Recently, graph regularization has also been used in conjunction with AEs to preserve the geometric spatial structure of HSIs [44], [45], [46], [47]. In addition, more and more features are mined to distinguish anomalies from the background. This can be accomplished using different strategies, i.e., combining CNNs and AEs to simultaneously capture spatial and spectral information for hyperspectral anomaly detection [48], [49], [50], [51], extracting both low- and high-frequency features to detect anomalies [52], collecting local patch and interband correlation information for improved performance [53], and so on.

Immediately following the AE-based methods, GAN-based anomaly detectors assume that the background obeys a predefined distribution to generate the corresponding data by a generator, and then a discriminator is built to judge whether the data come from a real dataset or a generator, until the discriminator cannot distinguish between them. In this case, the reconstructed data from the generator only contain background information. In [54], the Gaussian distribution function is adopted to produce fake data to force the encoded feature to obey the Gaussian distribution. In [55], spectral constraint and adversarial loss are designed with reconstruction loss to address the spectral variability of HSIs. In addition, weakly supervised detectors are also proposed to use some background or anomaly information. In [56], some background samples are first obtained by an unsupervised background searching method with the aim of training the network to capture background distribution. In [57], both background and anomaly samples are selected to get discriminative information to train the network and then combined with the low-rank model to conduct anomaly detection.

C. Motivations

Many research investigations have indicated that the traditional anomaly detection methods (statistical- and representation-based) are capable of solving anomaly detection tasks in a simple background in a satisfactory way by making some assumptions or building a pure dictionary. However, there is a great challenge for these traditional methods when

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high-dimensional HSIs are faced with real-world complex application scenarios. To solve this issue, deep learning offers a necessary and important solution (with impressive learning capability) to mine deep information from sophisticated cases. Among the existing deep-learning-based hyperspectral anomaly detection methods, the CNN-based detectors need to train the network with the help of a priori information and then conduct anomaly detection, which is not practical in real applications. Comparatively, unsupervised AE- and GAN-based detectors are more suitable for anomaly detection tasks. However, these methods are still in the preliminary exploration stage, and the following problems need to be solved.

1) Both the AE and GAN-based methods encode and decode features in units of the pixel so that spatial information is neglected. The CNN-based methods capture some local spatial information but are not spatially adaptive and are sensitive to the size of the target. Moreover, the external information of the convolutional kernel is ignored due to the small perception field.
2) All the above methods detect anomalies by reconstructing the background, but experiments show that most of them inevitably reconstruct anomalies at the same time, which is especially obvious in large target detection scenarios in complex backgrounds. Therefore, how to suppress the expression of anomalies is an important issue.
3) Hyperspectral anomaly detection networks based on AEs include encoding and decoding stages. Most of the existing detectors only impose various constraints on the encoded features to obtain implicit features of the background. However, it is practically shown that the implicit features inevitably contain anomaly information, and how to weaken the anomalies in the decoding stage for efficient background reconstruction has not been well-studied.
4) There is no basis for anomaly detection network design (such as the number of coding and decoding layers, the number of filters, and filter size) As a result, the construction of the network structure is purely handmade, becoming an exercise of trial and error that disregards the physical meaning.

D. Proposed Contribution

To address the above issues, a new background-guided deformable convolutional AE (DCAE) is designed for hyperspectral anomaly detection. DCAE is mainly composed of three mutually supportive parts, i.e., encoder, decoder, and background guidance module. The encoder includes two identical spatial information extractors and an external information extractor. The former captures adaptive and realistic spatial structures by combining regular convolution and deformable convolution (DC), while the latter introduces the external information of the convolution kernel using a nonlocal convolution (NLC). More importantly, a well-designed mask is embedded in the NLC to suppress high-frequency anomalous information. The decoder design originates from the idea of the linear reconstruction model of traditional methods, i.e., the reconstruction process is regarded as a linear combination of the encoded vector and the weight of the decoding layer. And a background information extractor module is designed to guide the proposed network learning. Indeed, two guidance ways are proposed: one is to initialize the weight of the decoder, and another is to add a new loss term. Note that the decoder structure and the number of channels of the encoder are determined by the designed background guidance module, i.e., the encoder, the decoder, and the background guidance module support each other to realize the anomaly detection task. The main innovative contributions of this work can therefore be summarized as follows.

1) To fit the complex spatial structure, in the encoder, the spatial feature extractor introduces DC into the regular convolution to enhance the network ability to extract irregular information. Furthermore, NLC is introduced for the extraction of information outside the convolution kernel. The combination of DC and NLC uses both the local and global properties of the HSIs.
2) To remove the anomalous information in the encoder stage, a mask-embedded NLC is proposed. In the reconstruction network, the attention mechanism focuses on the high-frequency information part of the image, i.e., the anomalies of hyperspectral anomaly detection. Therefore, the designed mask hides high-frequency information from the attention graph, making the network focus more on widespread backgrounds than on anomalies.
3) Most importantly, to enhance the decoder capability to reconstruct the background, during the decoder design (and inspired by linear reconstruction), a background guidance module is designed (in two ways) to guide the proposed network to reconstruct the background well. Meanwhile, the background guidance module directly determines the design of decoding layers and the number of filters in the hidden layer, which adaptively adjusts to different datasets. As far as we know, this is the first time in the literature that the focus is on the decoding stage optimization for better anomaly detection performance.

E. Paper Organization

The remainder of this article is organized in five sections. Section II introduces the DC and NLC briefly. Section III presents the conceptual design and implementation details of the proposed DCAE, including the network framework, the adaptive spatial feature extractor, the external information extractor, the background guidance module, and the loss function. Section IV shows comparison of the anomaly detection results obtained by the proposed method with those obtained by other eight competitors on seven real datasets, using both quantitative and qualitative criteria. Section V delves into the performance of the proposed network through a number of ablation experiments and some toy experiments. Finally, Section VI concludes this article with some remarks and hints at plausible future research lines.

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II. Related Works

This section provides the key elements in the design of the proposed model, i.e., the DC and the NLC.

A. Deformable Convolution

The adaptivity of the DC is accomplished by learning an offset [58]. Specifically, taking a 3×3 kernel as an example, the mathematical expression of the original convolutional process can be written as follows:

$$y(P_0) = \sum_{P_n \in \Re} w(P_n) * x(P_0 + P_n)$$
$$\Re = \{(-1,-1), (-1,0), \ldots, (0,1), (1,1)\} \quad (1)$$

where y is the data after convolution, x is the original data, P_0 is the location of the center pixel in the convolutional kernel, P_n is the location of nine pixels referenced to the center pixel in the convolutional kernel following the regular grid $\Re$, and $w(P_n)$ is the corresponding weight of the pixel at P_n. Equation (1) means that the feature value out of a convolution is the weighted sum of all the pixels in the convolutional kernel. Compared with the regular convolution process, the DC will add another parameter to learn as follows:

$$y(P_0) = \sum_{P_n \in \Re} w(P_n) * x(P_0 + P_n + \Delta P_n) \quad (2)$$

where ΔP_n is the offset parameter of all the sampling locations in the convolution kernel. Generally, the offset parameter is learned by comparing the original feature map with the feature map after one convolution.

B. Nonlocal Convolution

In computer vision, to capture the long-range dependencies without increasing network depth, the nonlocal neural network was proposed [59]. Unlike the traditional convolution, NLC considers the relationships among pixels in an image from a global perspective, rather than just considering the local spatial relationships among neighboring pixels. It works by computing a weighted sum of all the pixels in an image, where the weights are determined based on the similarity between each pixel and the current pixel being processed, specifically

$$y_i = \frac{1}{\boldsymbol{C}(\mathbf{X})} \sum_{\forall j} f(x_i, x_j) g(x_j) \quad (3)$$

where $\mathbf{X}$ is the feature map or input image, i and j give the location of the pixel, f computes the similarity of the pixel x_i with other pixels in the feature map, g is the value of x_j in another feature map, and C is a generalization function. When using NLC, f and g have different choices. Taking Gaussian embedding as an example, i.e.,

$$f(x_i, x_j) = e^{\theta(x_i)^T \varphi(x_j)}, \quad g(x_j) = W_g x_j \quad (4)$$

where $\theta(x_i)$ and $\varphi(x_j)$ are two embedded terms, and W_g is the weight matrix to be learned.

III. Proposed Methodology

Our newly proposed background-guided DCAE comprises an encoder, a decoder, and a background guidance module. Among them, the adaptive spatial features' extractor (in the encoder), the external information extractor (in the encoder), and the background guidance module are the key contents. In this regard, the methodology is described with a focus on the following five terms: 1) the framework of DCAE; 2) the adaptive spatial feature extractor; 3) the external information extractor; 4) the background-guided network learning; and 5) the loss function. Each term is described in further detail below.

A. Framework of DCAE

To extract deep spatial and spectral features, the convolutional AE is used as the basic component to design the DCAE network. The overall diagram of the proposed method is shown in Fig. 1, which is mainly composed of three parts, convolutional encoder, convolutional decoder, and background guidance module. Specifically, the encoder includes three layers, the decoder has only one layer, and the background guidance module is designed to promote the generation of the decoding feature map. In particular, the design of the decoding and encoding layers is closely linked to the background guidance module, and the three support each other to form the proposed network.

Regarding the encoding stage, the first two encoder layers are adaptive spatial feature extractors with the same structure (regular convolution kernel combined with the DC kernel) except for the number of filters. Let us assume that the size of the input image is $\mathbf{Y} \in \mathbb{R}^{B \times H \times W}$, where B is the number of bands, and H and W are the height and the width of the image, respectively. In the two encoder layers, a 3×3 regular convolutional kernel is first adopted for 2-D convolution followed by the LeakyReLU activation function, 2-D BatchNorm, and dropout. The number of filters of the two encoder layers is $B/2$ and $B/4$, respectively. Then, 2-D DC with the SoftMax activation function is introduced to adjust the shape of the regular kernel and capture flexible spatial information. The third encoder layer is designed to extract external information of the local convolution kernel. Thus, an NLC with a well-designed mask is implemented following a 2-D convolution layer with 1×1 kernel and the SoftMax activation function, to further emphasize background information and weaken anomalous information. Meanwhile, according to the idea of linear reconstruction, the background module is constructed to guide the output of the decoder. Thereby, the decoder layer must recover the original image directly from the encoded vector, i.e., using one 2-D convolution layer in the decoder with a 1×1 convolution kernel but no activation function. It should be noted that padding is used to keep the size of the feature map constant throughout the convolution process.

B. Adaptive Spatial Features' Extractor

In HSIs, the distribution of ground objects is sophisticated and the shape of targets is different and uncertain.

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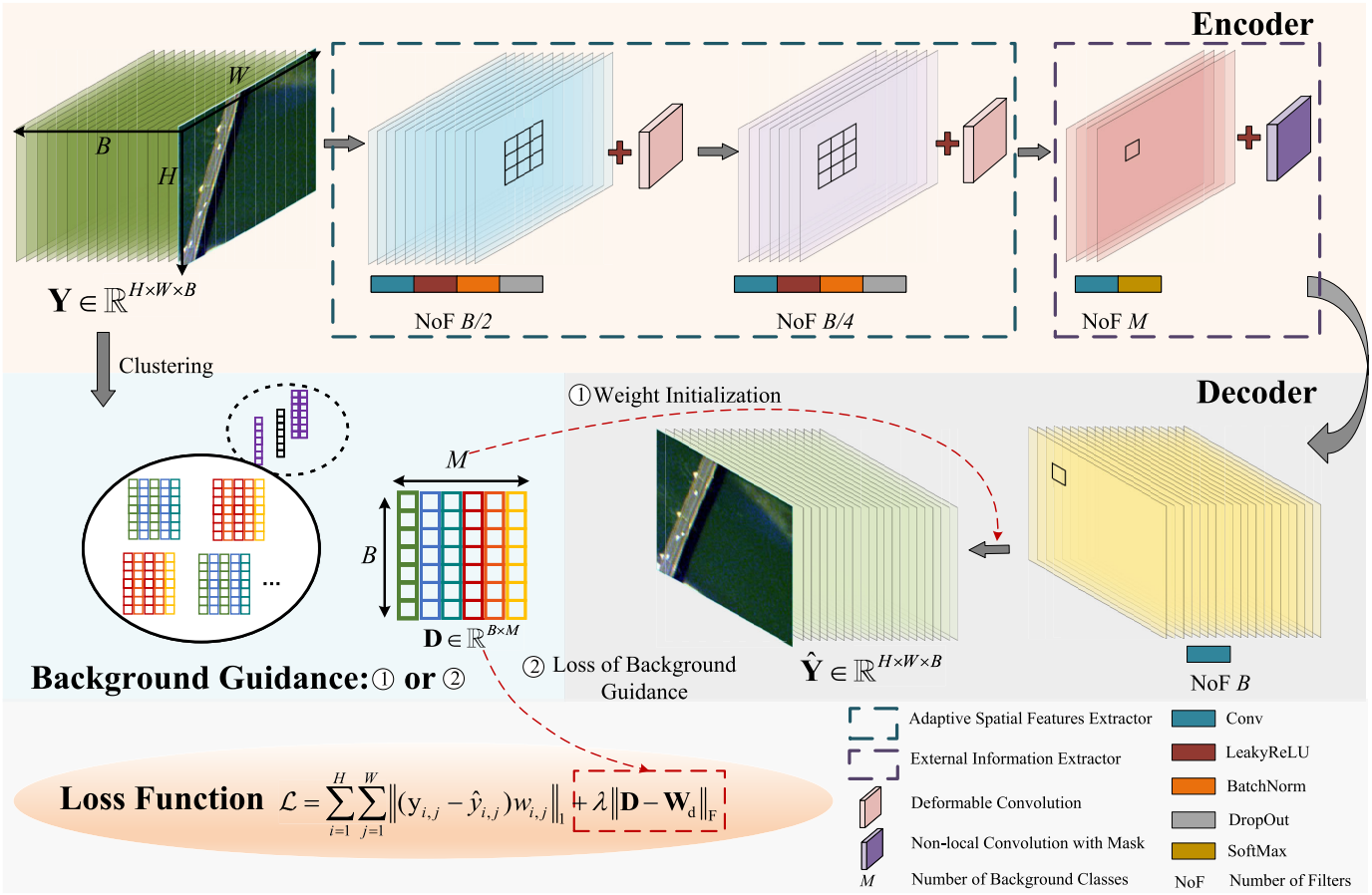


Fig. 1. Schematic of the proposed DCAE method. Two versions are implemented according to the adopted background guidance strategy, as described in Section III-E: DCAE with weight initialization (DCAE−I) and DCAE with added loss terms (DCAE−L).

To harmonize this issue, in the traditional anomaly detection methods, super-pixel segmentation, spatial clustering, and adaptive shapes are used to extract effective features. In deep-learning-based anomaly detection methods, although super-pixel segmentation has been adopted in some studies to construct a graph to capture flexible spatial relationships among the pixels, it is only designed for AE-based networks. The CNN AE-based networks generally use a regular convolutional kernel to extract local information, which needs to be resized according to different images. In this context, the convolutional kernel with a fixed size fails to match the features in real complex scenes, reducing the generalization capability of the network. To overcome this issue, we combine DC presented in Section II with regular convolution, i.e.,

$$\begin{aligned}\mathbf{H}_1 &= \mathrm{DeformConv2d}(\mathrm{Conv2d}(\mathbf{Y}))\\ \mathbf{H}_2 &= \mathrm{DeformConv2d}(\mathrm{Conv2d}(\mathbf{H}_1))\end{aligned} \tag{5}$$

where $\mathbf{H}_1$ and $\mathbf{H}_2$ are the hidden feature maps of the first and second layers in the encoder, respectively. Fig. 2 shows that the regular convolution may introduce chaotic information to the convolutional feature map, while each sampling of DC is more adaptive and similar to real scenarios. In the case of anomaly detection, DC can help compress the background and anomalous information better during the encoding stage.

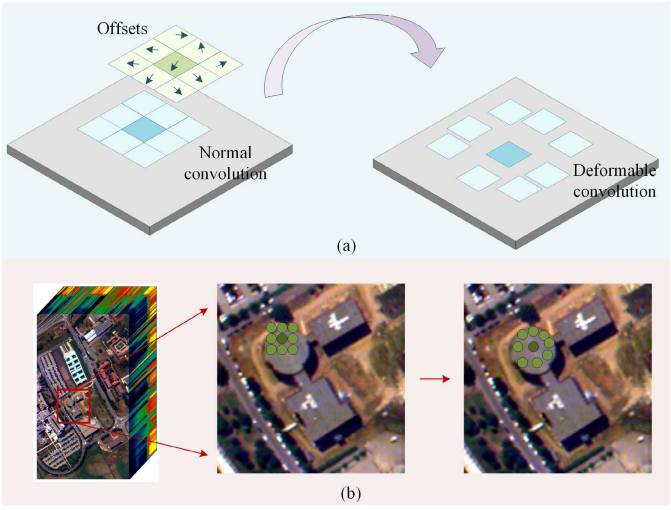


Fig. 2. Diagram illustrating the process of DC. (a) Traditional convolution to DC. (b) Real example.

C. External Information Extractor

Due to the huge disparity in the number and the distribution of backgrounds and anomalies, global information is helpful and important to separate backgrounds and anomalies. In the existing AE-based detectors, graph construction or statistical distribution is normally used to estimate the global relationship

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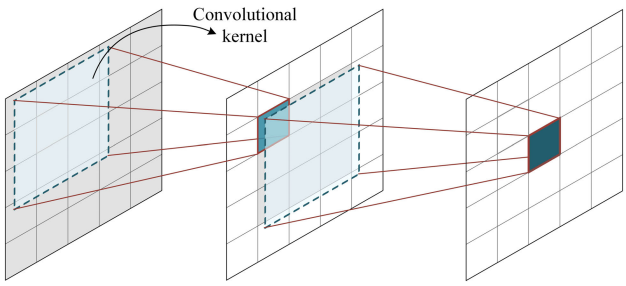


Fig. 3. Receptive field of regular convolution, taking an image of size 5×5 with a 3×3 convolutional kernel as an example.

among pixels for anomaly detection purposes. Nonetheless, in the existing hyperspectral anomaly detection methods based on CNNs, global information is neglected. These methods only adopt a convolutional kernel or spatial patch to integrate local spatial information. Although the global information can be captured by adding the number of convolutional layers, it requires increased running time and computation to find the optimal parameters. For example, as shown in Fig. 3, when a 3×3 convolutional kernel is used for an image of size 5×5 with nonpadding and a stride of 1, a two-layer network is required to gain global information. In this way, when using the same convolution kernel with an image of size 100×100, at least 44 convolution layers are needed to gain the global information, which implies a significant number of parameters to be trained (overfitting problems) arranged over a large depth (gradient vanishing problems). Thereby, how to subtly extract the external information of the convolution kernel is crucial for hyperspectral anomaly detection purposes.

The global information extraction capacity of NLC is introduced into the proposed network. It can be seen from Section II-B that NLC is similar to the self-attention mechanism [60]. θ, φ, and g are three mapping functions to different feature spaces. In this sense, $f(x_i, x_j)$ measures the similarity among the pixels of $\mathbf{X}$ in two different feature spaces, i.e., the similarity computation of self-attention. The product of f and g is the weighted sum considering the information of all the pixels. In other words, NLC is a feature extraction process considering all the pixel information, while regular convolution is a feature extraction process that only considers the pixel information inside the convolution kernel.

In terms of hyperspectral anomaly detection, if a nonlocal CNN is adopted directly, two problems will arise. The former is that the objective of an AE-based anomaly detection network is to reconstruct backgrounds rather than anomalies, i.e., the network should be designed with this intention in mind. The latter is that the computation of NLC is time-consuming if the tested image is large. To solve these two potential issues, two strategies are designed based on NLC for the hyperspectral anomaly detection task.

1) Mask for Nonlocal Convolution: In the reconstruction network, the NLC improves the performance of image reconstruction. Nevertheless, the recovery of anomalous pixels is not expected in anomaly detection tasks, which are supposed to be detected by reconstruction errors. A careful analysis of the implementation of NLC shows that this self-attentive model is prone to "shortcut" problems. Specifically, the output of the model is likely to be a replica of itself, without considering other information, resulting in anomalies that can be well-reconstructed. Moreover, the anomalous targets in an image are likely to be multiple; thus, it is necessary to prevent other similar anomalous information from contributing to the recovery of the anomalous targets. In this work, a mask is designed to alleviate this problem. Specifically, the mask is designed for the similarity matrix between θ and φ, which cuts off the representation of similar pixels in φ to the PUT in θ. As shown in Fig. 4, the similarity between each PUT in θ and all the pixels in φ is computed first, and then the first 1% of the values with high similarity for each PUT are assigned to negative infinity. After that, the SoftMax activation function is used to make the influence of those pixels with high similarity to the current PUT to be zero. The function expression is

$$S = \text{SoftMax}(\text{Mask}(\theta(\mathbf{H}_2)^T \varphi(\mathbf{H}_2))) \tag{6}$$

where $\mathbf{S}$ is the masked similarity matrix. And then the masked NLC (MNLC) is

$$\mathbf{H}_3 = \frac{1}{\boldsymbol{C}(\mathbf{H}_2)} e^S g(\mathbf{H}_2) \tag{7}$$

where $\mathbf{H}_3$ is the output feature of the encoder, and $\mathbf{H}_2$ is the output of the second layer of the encoder, respectively. In Fig. 4, the first two higher values in the similarity matrix are masked. Thus, for the anomalous target, the mask can eliminate the anomaly itself and other similar anomaly information to reconstruct the anomalies; for the background, the mask will only have a negligible effect on the effect of the background reconstruction because of the large number of background pixels.

2) Downsampling for Nonlocal Convolution: In NLCs, large matrix multiplications and convolutions are used several times. If the input image is relatively large, then the computational effort grows dramatically. To reduce the computational burden, downsampling is considered for matrices φ and g, conducting dimensionality reduction. At the same time, the spectral information of the input image is also extracted, and the number of features becomes half of the original one. Here, 2-D MaxPooling is adopted for downsampling. Thus, the function expression of the NLC is rewritten as

$$\begin{aligned} S &= \text{SoftMax}(\text{Mask}(\theta(\mathbf{H}_2)^T \text{MaxPool2D}(\varphi(\mathbf{H}_2)))) \\ \mathbf{H}_3 &= \frac{1}{\boldsymbol{C}(\mathbf{H}_2)} e^S \text{MaxPool2D}(g(\mathbf{H}_2)). \end{aligned} \tag{8}$$

The complete implementation of the external information extractor is shown in Fig. 5.

D. Background-Guided Network Learning

Different from target detection, anomaly detection is an unsupervised task without any prior knowledge of backgrounds and anomalies. Hyperspectral anomaly detectors based on AEs are carried out by assuming that the reconstruction errors of anomalous pixels are larger than those of normal pixels due to their small area and different spectra. The loss function is the reconstruction residual between the original image and the reconstructed image. But the network may fail

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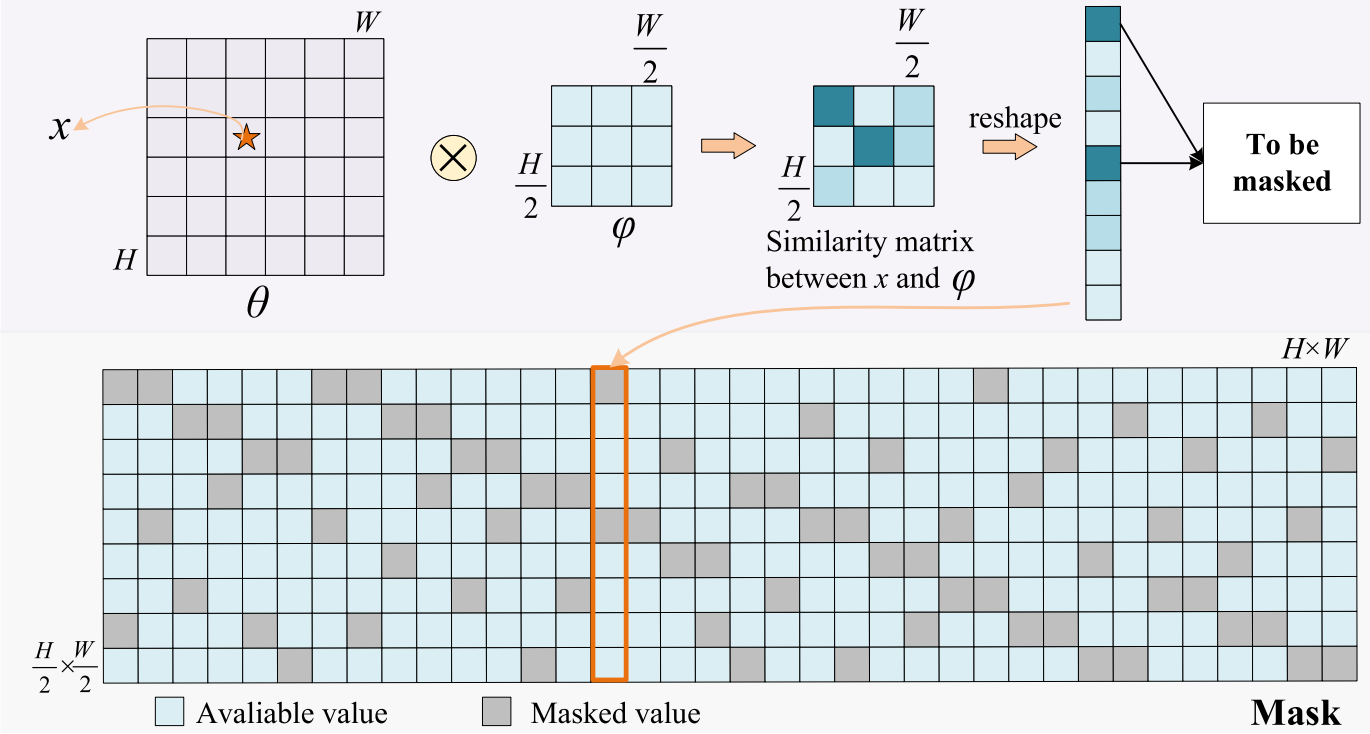


Fig. 4. Diagram illustrating how a mask for NLC operates. The similarity between each x in θ and all the pixels in φ is computed first, and then the first 1% of the values with high similarity for each PUT are assigned to negative infinity. In the figure, the first two higher values in the similarity matrix are masked.

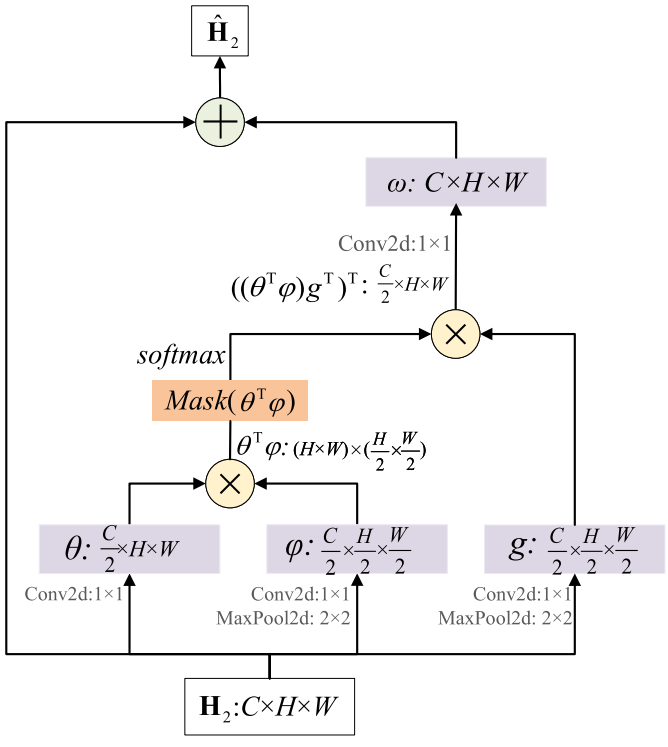


Fig. 5. Flowchart of NLC with the designed mask, in which C is the number of input channels.

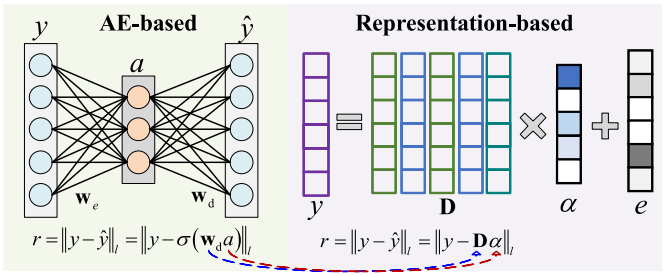


Fig. 6. Schematic of (left) AE-based and (right) representation-based anomaly detectors.

for large target images with a complex background, as the information of targets can also be well-reconstructed while the complex background cannot. To mitigate this problem, an external information extractor with a well-designed mask is designed in the encoding phase of the proposed network to get a good feature map from the encoder. Nonetheless, AE-based anomaly detectors include two stages, i.e., encoding and decoding. The decoding stage is as important to the image reconstruction results as the encoding stage. Thereby, how to constrain encoding and decoding simultaneously to generate the desired reconstructed image is a key issue.

Essentially, the principle of AE-based detectors is the same as that of representation-based detectors, as shown in Fig. 6. In AE-based detectors, the PUT is recovered by decoding an abstract feature which is obtained by encoding the original pixel. On the contrary, in representation-based detectors, the PUT is reconstructed by the linear representation of a dictionary, which is extracted from the original image. To avoid anomaly contamination, the former tries to generate an abstract feature map containing background information, and the latter tries to construct a pure background dictionary. In this sense, the two are similar, where the only difference is that the former

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uses a nonlinear activation function. Thereby, in this study, the possibility of introducing concepts from representation-based methods into AE-based detectors has been considered to guide the proposed network learning.

In particular, the decoding process of AE-based detectors is analogized to the representation process of representation-based detectors. As shown in Fig. 6, the dictionary $\mathbf{D}$ and the representation coefficient $\boldsymbol{\alpha}$ in representation-based detectors are equivalent to the weight $\mathbf{w}_d$ and the encoded features $\mathbf{a}$ in the decoding stage of AE-based detectors, respectively. Thus, for this purpose, the decoder in the proposed network is only composed of one convolutional layer with a 1×1 convolutional kernel without activation function, and the background vectors are adopted to guide the proposed network learning. Specifically, two guidance strategies are designed.

1) The background vectors are directly adopted to initialize the weight $\mathbf{w}_d$ of the decoder, i.e.,

$$\mathbf{D} \rightarrow \mathbf{w}_d. \tag{9}$$

2) So far, the deep learning network is a black box and, hence uninterpretable. Although the weight of the decoding stage is initialized, the subsequent update process is unpredictable and uncontrollable. Therefore, another guidance way is introduced by adding a loss function term, which is

$$\mathcal{L}_B = \|\mathbf{D} - \mathbf{w}_d\|_F. \tag{10}$$

By introducing the background guidance module, two significant advantages are gained. One is that the background can be better reconstructed by facilitating the generation of the ideal encoding vector $\mathbf{a}$ and the reconstructed vector $\hat{\mathbf{y}}$ simultaneously. The other one is that the convergence of the model can be accelerated.

Then how to obtain the background information is a key point. Representation-based anomaly detectors have studied all kinds of dictionary construction methods for this purpose, including clustering, endmember extraction, and dictionary learning. Among them, clustering methods have been widely used due to their simplicity and the fact that they are well-developed. In this article, to test the validity of the proposed idea, the classical K-means clustering method is also used to extract background information. Moreover, the elbow method, i.e., the sum of the squared errors (SSE), is used to determine the number of clusters. In the clustering method, the cluster centers of classes in which the number of pixels is more than 1% of the number of pixels in the image are selected as the background information. Accordingly, the number of filters for the last hidden layer is set as the number of dictionary atoms as shown in Fig. 1.

E. Loss Function

The AE-based anomaly detectors adopt the reconstruction error of the input image and the reconstructed image as the loss function, which is the mean square error (mse). In convolutional AE-based detectors, the same loss function is designed using the Frobenius norm of a matrix. Nonetheless, an interesting point should be analyzed, i.e., what is the meaning of the Frobenius norm of a matrix? In mathematics, the Frobenius norm of a matrix is the arithmetic square root of the sum of squares of all the elements in the matrix. Minimizing the Frobenius norm of a matrix means that all the elements of the matrix are desired to be small. In this way, the reconstruction error of an image is more driven by noise than anomalies. But the goal of hyperspectral anomaly detection is to detect anomalies in the image. Intuitively, the reconstructed residuals are expected to have values only for the anomalous parts and 0 for the remaining (i.e., sparse) parts. Moreover, the loss function is hoped to be insensitive to outliers, making the network more concerned with the background. In this sense, the l_1-norm is considered

$$\mathcal{L}_{l_1} = \left\|\mathbf{Y} - \hat{\mathbf{Y}}\right\|_1. \tag{11}$$

Furthermore, the adaptive weight optimization strategy in [48] is also used, i.e.,

$$\mathcal{L}_{l_1} = \sum_{i=1}^{H}\sum_{j=1}^{W}\left\|(y_{i,j} - \hat{y}_{i,j})w_{i,j}\right\|_1 \tag{12}$$

where $w_{i,j}$ is the weight of the pixel $y_{i,j}$ computed by reconstruction error, which is updated every 50 epochs. The proposed method implements two versions, one is `DCAE` with weight initialization (`DCAE-I`) and the other is `DCAE` with added loss terms (`DCAE-L`). Equation (12) is the loss function of the former, and the loss function of the latter can be expressed as

$$\mathcal{L} = \mathcal{L}_{l_1} + \lambda\mathcal{L}_B \tag{13}$$

where λ is the tradeoff between the reconstruction loss and the background-guided loss. Equation (12) or (13) is used to induce network learning until the convergence condition is reached.

IV. Experiments

To demonstrate the anomaly detection performance of the proposed methods `DCAE-I` and `DCAE-L`, this section uses seven real HSIs and eight comparison methods to conduct experiments. In the following, we discuss the hyperspectral datasets, experimental settings, and detection results.

A. Experimental Datasets

The details of the seven real HSIs (three small and four large scenes) used in this study are as follows.

1) The first dataset (**HYDICE Urban**) is a classical hyperspectral anomaly detection scene, acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) in 1995, with 1.56-m spatial resolution and 10-nm spectral resolution [61]. The pseudocolor and anomaly reference map are shown in Fig. 7-I. The scene covers an urban area and has a size of 80×100 pixels and 175 bands. The anomalies are some roofs and cars covering 21 pixels. Such anomalies are relatively small and even appear at the subpixel level. According to the elbow method, 5 is set as the number of clusters of the K-means clustering method.

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2) The second dataset (**Pavia Center**[1]) is a scene of a beach including a bridge, bare soil, water, and some vehicles on the bridge as shown in Fig. 7-II. The imaging sensor is the Reflective Optics Spectrographic Imaging System (ROSIS), and the spatial resolution of the scene is 1.3 m. In the selected image (of size 105 × 120 pixels), after removing the noisy bands, 102 effective bands are preserved to detect the vehicles on the bridge, which are regarded as anomalies. Regarding the number of clusters, 3 is selected by the elbow method for this dataset.
3) The third dataset (**Hyperion**) captures an agricultural region located in the State of Indiana, USA, in 2008 and it was collected by the Hyperion imaging sensor with 10-nm spectral resolution and 10-m spatial resolution. In the experiment, a 150 × 150 subimage with 155 available bands is cropped for anomaly detection purposes. As shown in Fig. 7-III, anomalies originate from both the storage silo and roofs, which visually blend into the surrounding background. The cluster number is determined as 5 from the elbow map.
4) The fourth dataset (**Gulfport**) recorded the scene of an airport in Gulfport, USA [62], and it was taken by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). The spatial resolution of the image is 3.4 m and the time of collection was July 7, 2010. As shown in Fig. 7-IV, the targets in this dataset are three planes with different sizes, which are almost unrecognizable to the naked eye in the false-color image, and the background is relatively simple. The whole image contains 100 × 100 pixels and 191 bands after removing the noisy bands. The number of clusters selected in this case is 4.
5) The fifth dataset (**Los Angeles**) is taken at the Los Angeles airport, USA, and it is composed of 100 × 100 pixels and 205 bands. Among them, two planes are regarded as anomalous targets. It was collected by the AVIRIS sensor and the spatial resolution is 7.1 m, while the background is more complex for these data. As indicated in Fig. 7-V, 4 is also a suitable choice for the number of clusters in this dataset.
6) The sixth (**San Diego I**) and seventh (**San Diego II**) datasets are two classical scenes of an airport taken by the AVIRIS sensor in San Diego airport, CA, USA. The size of the original image is 400 × 400 pixels with 186 effective bands and 3.5-m spatial resolution. In this work, two subimages with 100 × 100 pixels are cropped to conduct the experiments, as shown in Fig. 7-VI and VII. Both contain three planes that can be regarded as anomalies. As shown in the elbow map, 4 and 6 are selected to be the number of clusters for these images, respectively.

B. Comparison Algorithms and Experimental Settings

To compare the performance of the proposed method, both the traditional methods and deep-learning-based methods are selected to conduct experiments. The traditional methods comprise statistics-based methods (**RX** [10] and kernel isolation forest-based hyperspectral anomaly detection method (**KIFD**) [14]) and representation-based methods (collaborative representation detector (**CRD**) [20], **LRASR** [23], and **PTA** [30]). The deep learning-based methods include graph autoencoder-based methods (robust graph AE (**RGAE**) detector [45] and guided AE detector (**GAED**) [63]) and autonomous hyperspectral anomaly detection network (**Auto-AD**) [48].

In the proposed methods, except for the hyperparameters, only one parameter for DCAE-L should be adjusted, i.e., the tradeoff λ of the background guidance term in the loss function. The selection range of λ is set as $\{1e^{0}, 1e^{-1}, 1e^{-2}, 1e^{-3}, 1e^{-4}, 1e^{-5}, 1e^{-6}\}$. For both DCAE-I and DCAE-L, the learning rate is set as $1e^{-2}$ with a decay rate of $1e^{-8}$. Adam optimizer and Kaiming initialization are used to train the networks.

Regarding the comparison methods, the parameters of KIFD, LRASR, GAED, and Auto-AD are set to the values suggested in the original papers. The size of the dual-window of CRD is set in the range {(15, 13), (15, 11), (15, 9), (15, 7), (15, 5), (15, 3)}, and the regularization parameter of CRD is set in the range $\{1e^{-6}, 1e^{-5}, 1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}, 1e^{0}\}$. The setting ranges of (μ, α, β, and τ) in PTA are set in the range $\{1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}, 1e^{0}, 1e^{1}, 1e^{2}, 1e^{3}, 1e^{4}\}$ except for *rank* which is set in the range 1–20. The regularization parameter of RGAE is also set in $\{1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}\}$. Due to the uncertainty of the detection results of some methods (particularly deep-learning-based methods), the optimal result of 20 running results for every method is presented for the seven experimental datasets. In all the experiments, the deep learning methods were implemented on a cluster of NVIDIA GPUs, GeForce RTX 3070 with an Intel core i7 processor. The software stack is the PyTorch framework, running on a Linux system Ubuntu 22.04. The traditional approach was run on MATLAB 2016b for experiments.

The performance of the tested methods for the seven considered datasets is evaluated by receiver operating characteristic (ROC) curve quantitative evaluation, which can portray the relationship between the detection rate and the false alarm rate using the anomaly detection map (gray map) and the reference map (binary map). The area under the ROC curve (AUC) is regarded as the accuracy indicator of anomaly detection. The larger the AUC, the higher the accuracy.

C. Experimental Results

By conducting a large number of experiments, the optimal anomaly detection performance of all the considered algorithms is presented in this section for the seven considered datasets. The assessment of detection performance is folded up into two aspects, i.e., qualitative and quantitative analyses. The former comprises the anomaly detection maps shown in Figs. 8 and 9, respectively. The latter depicts the ROC curves and computes AUC as the accuracy metric, displayed in Fig. 10 and Table I. A detailed analysis of the obtained results per dataset follows.

1) HYDICE Urban: Intuitively, from Fig. 8(a), the anomaly detection maps of the proposed methods (DCAE-I and DCAE-L) exhibit a closer resemblance to the reference map

[1]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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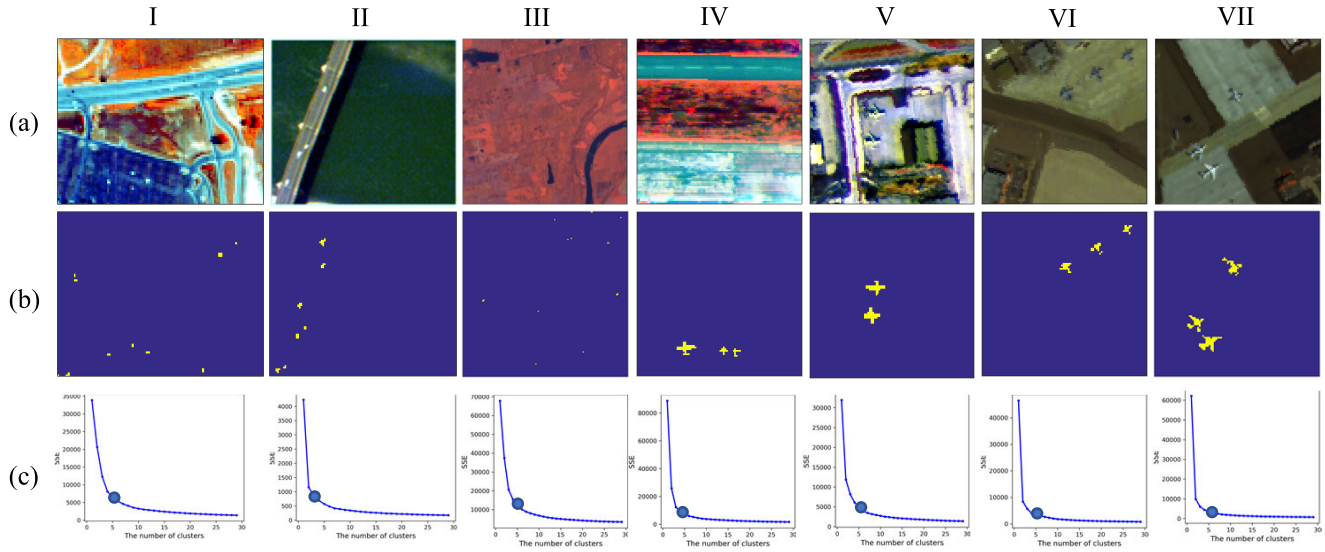


Fig. 7. Pseudocolor images and anomaly reference maps obtained from the considered experimental datasets. (I) HYDICE Urban. (II) Pavia Center. (III) Hyperion. (IV) Gulfport. (V) Los Angeles. (VI) San Diego I. (VII) San Diego II. (a) Pseudocolor images. (b) Corresponding reference maps with the true locations of anomalies. (c) Number of clusters estimated by the elbow method.

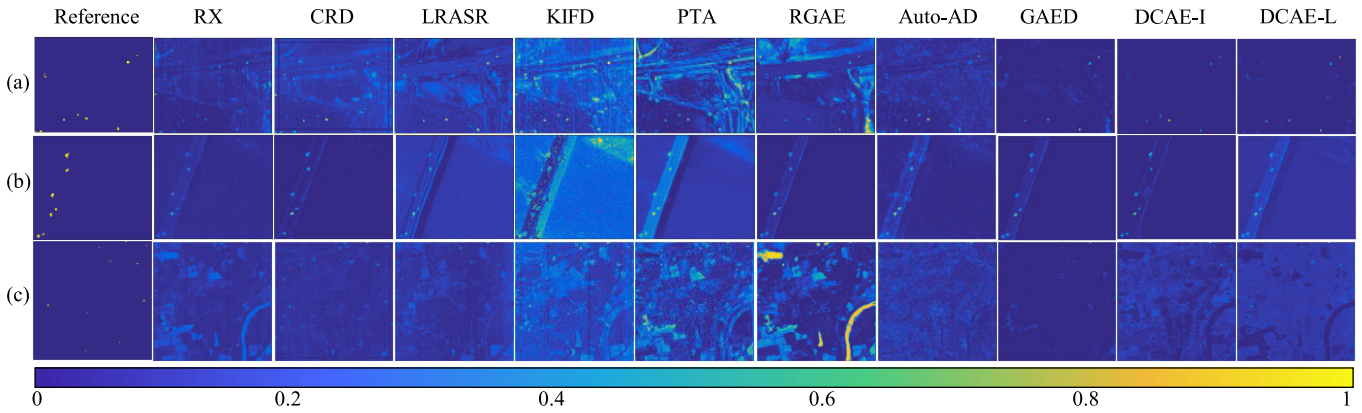


Fig. 8. Anomaly detection maps of all the considered methods for three small target detection datasets. (a) HYDICE Urban. (b) Pavia Center. (c) Hyperion.

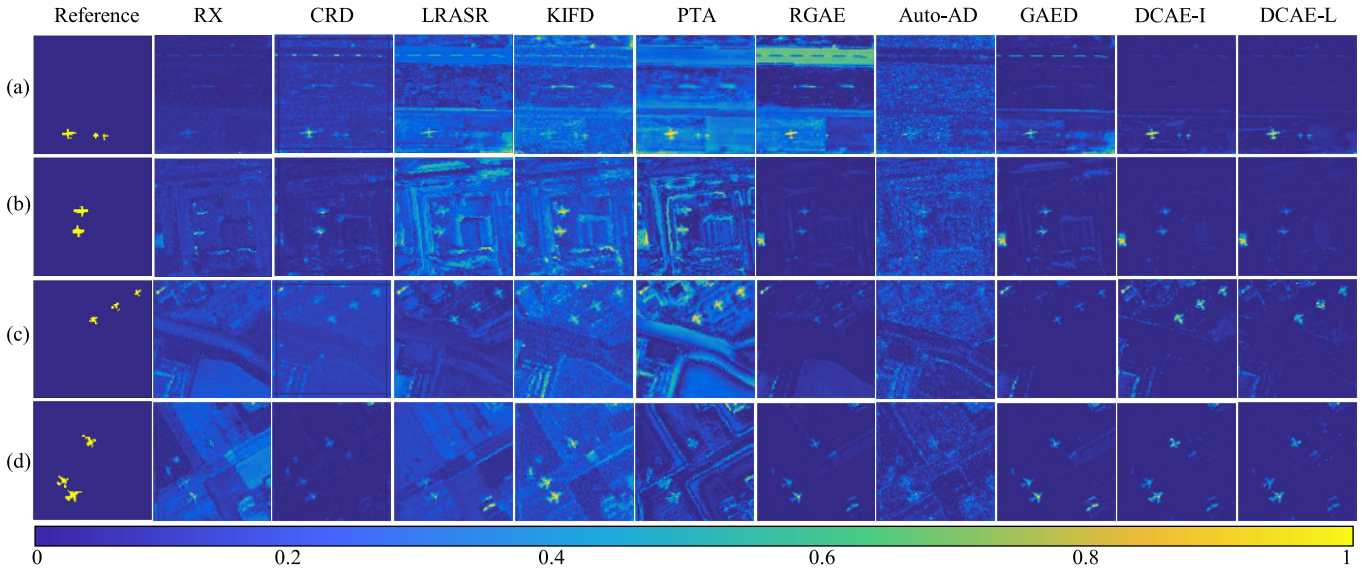


Fig. 9. Anomaly detection maps of all the considered methods for four large target detection datasets. (a) Gulfport. (b) Los Angeles. (c) San Diego I. (d) San Diego II.

than those provided by other methods. Remarkably, these two maps effectively capture targets, leaving a background that is pure enough to recognize anomalies with the naked eye. A detailed analysis in terms of overall accuracy, as shown in Table I, reveals that DCAE-I obtains the best accuracy, followed by KIFD, DCAE-L, and Auto-AD. According to the ROC curve in Fig. 10(a), the probability of detection of DCAE-I is higher than that of other methods, and the

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TABLE I
AUC AND RUNNING TIME OF SEVERAL ANOMALY DETECTION METHODS FOR THE SEVEN CONSIDERED DATASETS. BEST RESULTS IN BOLD AND SECOND-BEST RESULTS UNDERLINED

Dataset		RX	CRD	LRASR	KIFD	PTA	RGAE	Auto-AD	GAED	DCAE-I	DCAE-L
HYDICE Urban	AUC	0.9857	0.9499	0.9686	0.9972	0.9122	0.8128	0.9939	0.9663	**0.9984**	0.9963
	Time (s)	0.35	17.62	52.68	68.76	49.98	150.11	50.51	68.21	79.26	78.95
Pavia Center	AUC	0.9982	0.9537	0.9084	0.8463	0.9995	0.9995	0.9983	0.9994	**0.9997**	0.9997
	Time (s)	0.12	11.07	118.48	108.59	23.43	116.18	40.57	80.18	63.97	46.98
Hyperion	AUC	0.9227	0.9994	0.9975	0.9993	0.9844	0.8789	0.9962	0.9949	0.9992	**0.9994**
	Time (s)	0.41	22.40	138.91	171.74	101.75	380.20	20.51	195.83	79.57	109.42
Gulfport	AUC	0.9526	0.9917	0.9488	0.9792	0.9978	0.8890	0.9226	0.9690	0.9995	**0.9996**
	Time (s)	0.18	10.92	58.11	102.80	54.19	217.36	18.06	88.22	35.93	56.15
Los Angeles	AUC	0.8403	0.9779	0.8761	0.9785	0.9371	0.9579	0.8744	0.9680	**0.9884**	0.9838
	Time (s)	0.17	11.12	53.60	94.88	67.85	233.59	8.48	89.80	0.9937	0.9947
San Diego I	AUC	0.8545	0.9906	0.9880	0.9846	0.9916	0.9742	0.8726	0.9869	0.9937	**0.9947**
	Time (s)	0.17	10.82	59.20	117.61	57.82	193.28	14.12	91.85	87.77	91.48
San Diego II	AUC	0.7760	0.9786	0.8778	0.9917	0.9476	0.9891	0.9316	0.9884	**0.9945**	0.9934
	Time (s)	0.27	10.69	53.23	88.10	57.75	190.52	16.00	86.48	42.83	46.85
Average	AUC	0.8530	0.9774	0.9380	0.9681	0.9671	0.9288	0.9414	0.9818	**0.9962**	0.9952
	Time (s)	0.24	13.52	76.32	107.50	58.97	211.61	24.04	100.08	61.08	72.02

first one to get to 1 when the false alarm rate is negligibly small. Overall, the performance of all the considered methods can be summarized as: DCAE-I>KIFD>DCAE-L>Auto-AD>RX>LRASR>GAED>CRD>PTA>RGAE.

2) Pavia Center: Compared with the HYDICE Urban image, the scene of Pavia Center is relatively easy for anomaly detection tasks with larger targets and simpler backgrounds. In Fig. 8(b), both the traditional methods and deep learning methods exhibit good detection performance in visual terms (except for KIFD). Among them, the background suppression effect of DCAE-I, GAED, RGAE, and CRD is better than that of other methods. From the quantitative analysis, we can get similar conclusions. The performance of DCAE-I, DCAE-L, PTA, RGAE, and GAED is excellent. When the false alarm rate is less than 0.0002, the probability of detection of these five methods can approximately reach 1. The performance of all the methods can be summarized as: DCAE-I > DCAE-L>RGAE>PTA>GAED>Auto-AD>RX>CRD>LRASR>KIFD.

3) Hyperion: The anomalies in this dataset are not distinguishable visually from the surrounding background. In the anomaly maps of Fig. 8(c), the anomalous targets for all the methods are not visually obvious, and the anomaly maps for RGAE even show serious false positives. From the corresponding ROC curves of Fig. 10(c), it can be seen that the proposed DCAE-I and DCAE-L consistently outperform other methods when false alarm rates are low, followed by KIFD and CRD. According to Table I, the performance of all the methods can be summarized as: DCAE-L>CRD>KIFD>DCAE-L>LRASR>Auto-AD>GAED>PTA>RX>RGAE.

4) Gulfport: This dataset is a planar target detection scene for which both completeness and prominence of targets are important. From Fig. 9(a), it can be observed that the proposed methods DCAE-I and DCAE-L exhibit remarkable detection capabilities. Not only the background is suppressed but also the integrity and saliency of the anomalous targets are also highlighted. Comparatively, other methods cannot achieve such performance in both the aspects at the same time. Regarding ROC curves and AUCs, the performance of DCAE-I and DCAE-L is far better than that of other methods. In Fig. 10(d), the probability of detection of DCAE-I and DCAE-L is always much higher than that of other methods when the false alarm rate is in the range of 0–0.001, and the next is PTA. They both achieve an accuracy of more than 99.5%. The detection performance of all the methods for this dataset can be summarized as: DCAE-L > DCAE-I> PTA> CRD>KIFD>GAED>RX>LRASR>Auto-AD>RGAE.

5) Los Angeles: This scene is particularly challenging for anomaly detection purposes. The background is sophisticated and suspicious targets exist. In the anomaly maps depicted in Fig. 9(b), it is difficult to evaluate which algorithm works best in visual terms. The proposed DCAE-I and DCAE-L, GAED, and RGAE have a good background suppression effect, but they suffer from the interference of false targets. Conversely, KIFD, PTA, and CRD obtain good performance in highlighting anomalies but struggle to mitigate the impact of the conspicuous background. The rest of the methods exhibit subpar performance in both qualitative and quantitative terms. From the results of the quantitative experiments, the overall accuracy of the proposed method is the highest. In Fig. 10(e), the detection rate of the proposed method is consistently ahead of other methods for false alarm rates greater than 0.01, followed by KIFD and CRD. Although the accuracy of all the methods for this dataset is not ideal, the accuracy of the proposed methods can still approach 99%. The performance of all the methods for this dataset can be summarized as: DCAE-I> DCAE-L>KIFD> CRD>GAED>RGAE>PTA>LRASR>Auto-AD>RX.

San Diego I: Similar to the previous dataset, it is difficult to implement anomaly detection for this scene. Fig. 9(c) illustrates that PTA, KIFD, and LRASR struggle with

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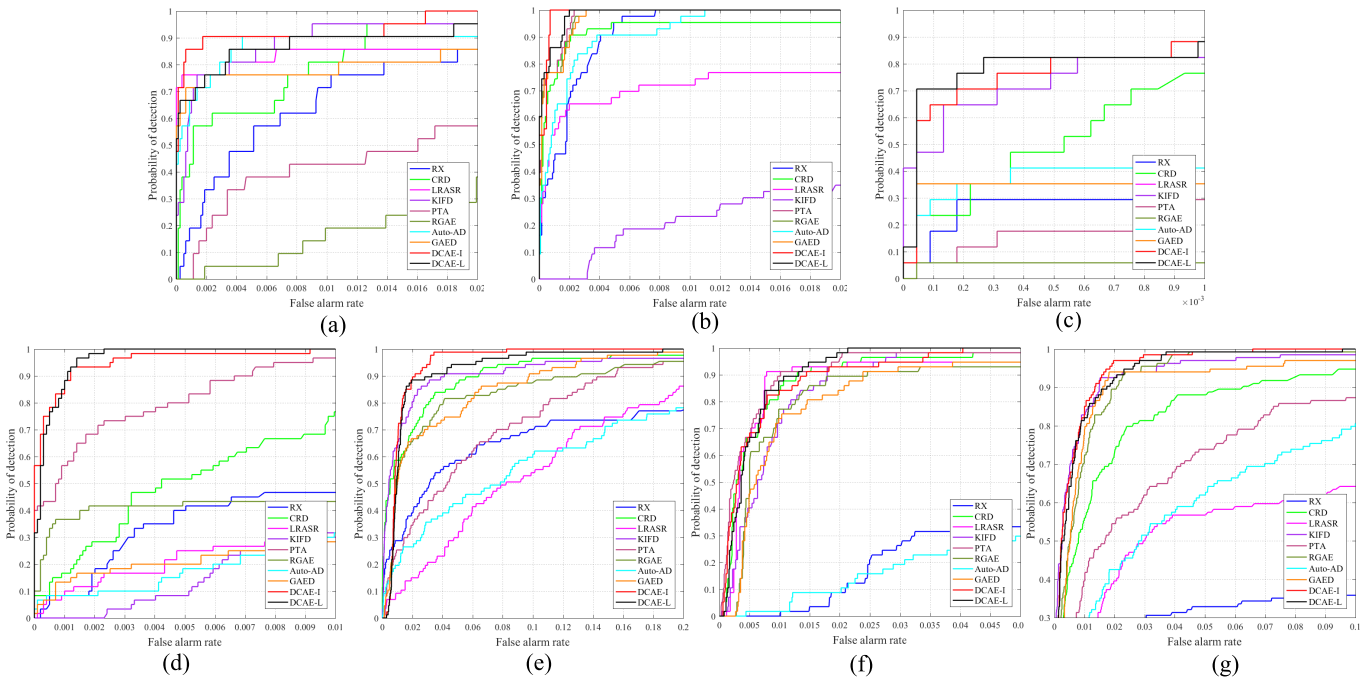


Fig. 10. ROC curves of all the considered anomaly detectors for seven experimental datasets. (a) HYDICE Urban. (b) Pavia Center. (c) Hyperion. (d) Gulfport. (e) Los Angeles. (f) San Diego I. (g) San Diego II.

background suppression, despite the saliency of targets in the anomaly map of PTA. However, in comparison to the experiments with the Los Angeles dataset, the proposed methods DCAE−I and DCAE−L achieve a notably superior target highlighting effect. Observing the ROC curves in Fig. 10(f), it is evident that RX and Auto-AD perform poorly, while the ROC curves for the other methods are less than satisfactory, and the advantages of the proposed method are not pronounced. But the AUC in Table I reflects that the accuracy of the proposed method surpasses that of other methods to some extent. Summarizing the performance of all the methods on this dataset, the ranking can be described as follows: DCAE−L>DCAE−I>PTA>CRD>LRASR>GAED>KIFD>RGAE>Auto-AD>RX.

San Diego II: In the visualized anomaly maps depicted in Fig. 9(d), the anomaly targets in RGAE, GAED, and the proposed DCAE−I and DCAE−L are more prominent and have superior background suppression than other methods. In terms of ROC curves and AUC, as shown in Fig. 10(f), KIFD and the proposed DCAE−I and DCAE−L exhibit superior detection performance compared with other methods when the false alarm rate is below 0.01. Furthermore, as the false alarm rate exceeds 0.04, the proposed DCAE−I and DCAE−L achieve an impressive detection rate, nearly reaching 1 while maintaining remarkable stability. The performance of all the methods for this dataset can be summarized as: DCAE−I>DCAE−L>KIFD>RGAE>GAED>CRD>PTA>Auto-AD>LRASR>RX.

Summary: The seven experimental datasets can be simply divided into two categories. The HYDICE Urban, Pavia Center, and Hyperion comprise small targets, while the other four datasets comprise large targets. From this point of view, it is obviously observed that the proposed methods can always perform well in scenes with small targets. If the background of the planar target detection scene is widely distributed and pure, the proposed methods can also obtain excellent results. Notwithstanding their satisfactory performance, the proposed methods may not be as effective in scenarios where some backgrounds possess anomalous features, such as the Los Angeles and two San Diego datasets. This is because the original intention of the proposed method is to detect the target of interest that accounts for the least percentage of the image. If the target of interest is not the least dominant in the image, then the network may not be optimal. Nevertheless, it is worth noting that our proposed methods significantly surpass other state-of-the-art approaches. In addition, the performance of the DCAE−I is usually better than that of the other version (DCAE−L), which indicates that the initialization of the hyperparameter is generally more important to the network than the corresponding loss term. In depth, the hyperparameter initialization determines the starting point of gradient descent and plays a decisive role in the final result obtained by the network.

Furthermore, careful observation reveals that the anomaly detection maps of RGAE and GAED are largely similar to those obtained by the proposed methods. In terms of design philosophy, they are similar for two reasons: they all use AEs as the basic framework of the network, and they all introduce global information to facilitate anomaly detection. However, they are totally different in terms of technology. RGAE and GAED introduce different graph regularizations into the AE to constrain the hidden feature to be ideal, but the proposed method not only introduces the DC and NLC but also designs a mask embedding in the NLC to highlight

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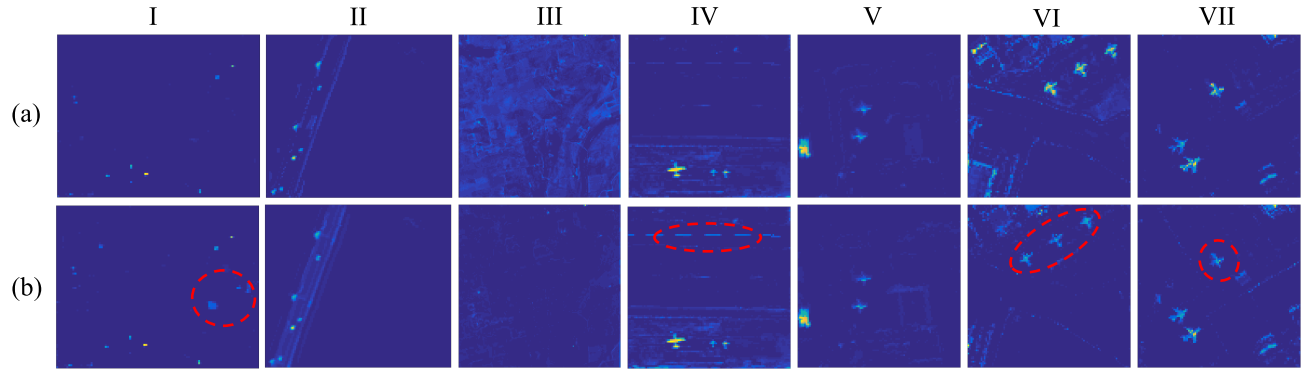


Fig. 11. Anomaly maps obtained using (a) DCAE−I and (b) DCAE−I without the mask. (I) HYDICE Urban. (II) Pavia Center. (III) Hyperion. (IV) Gulfport. (V) Los Angeles. (VI) San Diego I. (VII) San Diego II.

TABLE II
COMPARATIVE PERFORMANCE (AUC) OF THE PROPOSED METHODS AFTER CONDUCTING ABLATION EXPERIMENTS. (I) HYDICE URBAN. (II) PAVIA CENTER. (III) HYPERION. (IV) GULFPORT. (V) LOS ANGELES. (VI) SAN DIEGO I. (VII) SAN DIEGO II

Models	I	II	III	IV	V	VI	VII
DCAE-I	**0.9984**	**0.9997**	**0.9992**	**0.9995**	**0.9884**	**0.9937**	**0.9945**
w/o DC	0.9962	0.9995	0.9989	0.9972	0.9847	0.9881	0.9910
w/o MNLC	0.9973	0.9996	0.9979	0.9985	0.9860	0.9900	0.9919
w/o mask	0.9980	0.9998	0.9984	0.9992	0.9878	0.9903	0.9923

the anomalies. Moreover, the proposed method also designs a background-guided module to enhance the network and add some physical meaning. The experimental results indicate that the proposed strategies are more effective.

V. DISCUSSION

In this part, the performance of the proposed methods DCAE−I and DCAE−L is analyzed deeply, including ablation experiments, the impact of regularization on the loss function, and a parameter sensitivity analysis.

A. Ablation Experiments

1) What About the Adaptive Spatial Features' Extractor and the External Features' Extractor?: In the proposed networks, there are three novel architectural designs: the adaptive spatial features extractor, the external features extractor, and the well-designed mask for the external features extractor. It is worth exploring whether they are helpful for anomaly detection purposes or not. For convenience, ablation experiments are performed using DCAE−I as an example. The experimental results are summarized in Table II. According to Table II, apart from the Pavia Center dataset, the three strategies are obviously effective in both small and large target detection scenarios. Comparing DCAE−I with DCAE−I without DC, it can be seen that DC significantly improves the accuracy of HYDICE Urban and four large target detection datasets. As for DCAE-I without MNLC, the external information can be demonstrated to be useful for all the considered datasets. And the designed mask is the same. Regarding the Pavia Center dataset, the background is relatively simple and the targets are clear and small. Thereby, it is easy to detect anomalies so that the effect of the two extractors is slight. The experimental results indicate that the spatial feature extractor facilitates the preservation of the spatial structure of both the targets and background simultaneously and the external feature extractor, with the well-designed mask, can globally promote the background representation while suppressing anomalous information.

In Fig. 11, the impact of the designed masks is visually demonstrated. In the HYDICE Urban, Pavia Center, and Gulfport datasets, DCAE−I without masks exhibit some minor residual background in comparison to DCAE−I with the mask. In the two San Diego datasets, while some background areas are not perfectly reconstructed due to the presence of masks, the anomalous targets stand out more prominently. As for the Hyperion and Los Angeles datasets, the effect of the masks is imperceptible to the naked eye; however, it is evident from Table II that the mask continues to provide benefits.

2) Does the Background Guidance Work?: One of the contributions of this work is to design a background guidance module to promote the proposed network learning. To demonstrate the validity of the module, the proposed network without background guidance (called DCAE) is used to conduct experiments on seven datasets. In DCAE, the number of filters in the last layer of the encoder is set to a value that is the same as the number of identified clusters. A comparative performance assessment of DCAE, DCAE−I, and DCAE−L is shown in Fig. 12(h). It can be observed that the performance of DCAE is close to that of DCAE−L for the HYDICE Urban, Pavia Center, and Gulfport datasets. This is because, in a generalized sense, DCAE can be regarded as a special case of DCAE−L when the parameter λ of the background guidance term in the loss function approaches 0. In addition, the difference in detection accuracy of DCAE, DCAE−I, and DCAE−L is obvious in the HYDICE Urban, Hyperion, Los Angeles, and two San Diego datasets. Relatively speaking, the backgrounds of these five scenes are more complex than those of the other two, so this result suggests that the background representative atoms generated by clustering can assist background reconstruction to some extent.

Furthermore, the loss curves of DCAE, DCAE−I, and DCAE−L have been plotted as shown in Fig. 12(a)–(g). It is clear that the loss curve of DCAE−I is the fastest decreasing

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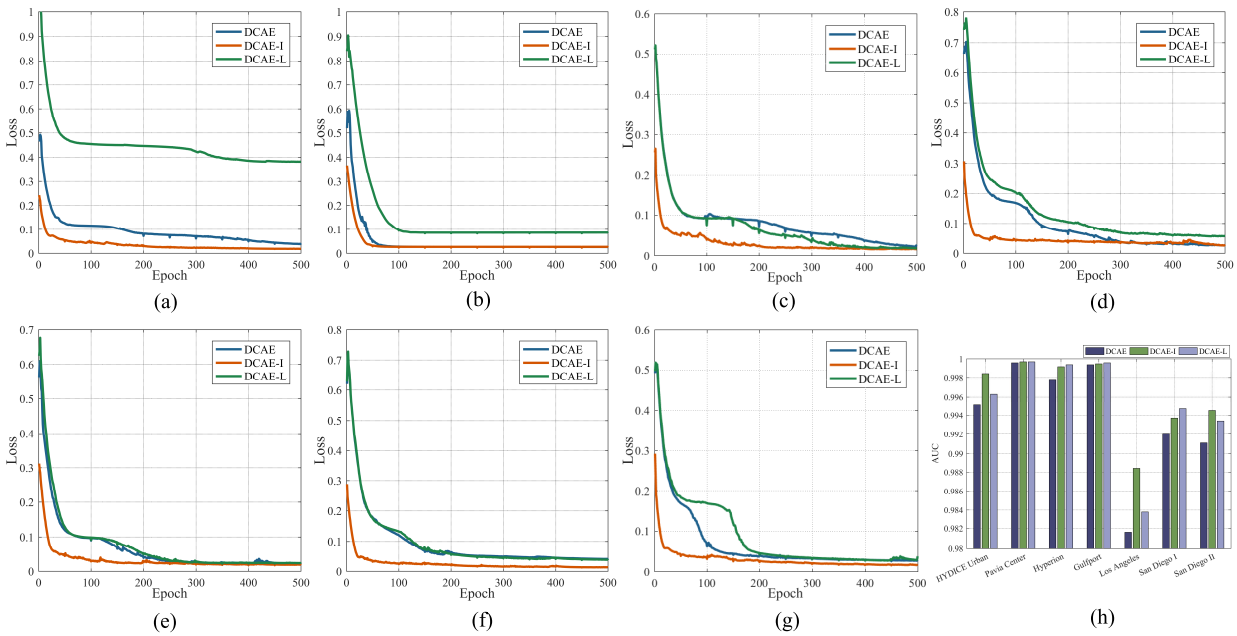


Fig. 12. Comparative performance of DCAE, DCAE−I, and DCAE−L for seven experimental datasets. (a) HYDICE Urban. (b) Pavia Center. (c) Hyperion. (d) Gulfport. (e) Los Angeles. (f) San Diego I. (g) San Diego II. (h) AUCs of the three methods for the seven datasets.

and most stable, much better than that provided by DCAE and DCAE−L. This is sufficient to show that the initialization of the decoding layer weights is crucial to the stability and accuracy improvement of the network, which has not been investigated in all previous studies. In the case of DCAE and DCAE−L, the parameter λ of the background guidance loss term has a huge impact on the loss drop of DCAE−L. In the HYDICE Urban and Pavia Center datasets, λ is set to $1e^0$ and $1e^{-1}$, respectively. Leaving aside the size of the loss, the decreasing curves of the loss of DCAE and DCAE−L are similar, as shown in Fig. 12(c)–(f). This reveals that the reconstruction residual loss still dominates the gradient descent direction, even with the addition of background guidance loss, when the initial weights are fixed. These observations, together with the AUCs in Fig. 12(g), lead us to conclude that the accuracy and stability of DCAE−I are excellent, while DCAE−L has only a slight advantage in terms of accuracy and is generally as stable as DCAE. Overall, our introspection is that the background guidance module is beneficial for the detection performance of the proposed network.

B. Which l_{norm} in the Reconstruction Loss Term Is Really Helpful for Hyperspectral Anomaly Detection?

In all the existing anomaly detection networks, the reconstruction loss term in the loss function adopts the l_2-norm. Nonetheless, from a physical sense, the l_2-norm of a matrix constrains the minimum of each value in the matrix, which is more consistent with noise. The anomalies are sparse rather than uniformly distributed in the matrix, so constraints such as the sparse norm l_1 and smooth l_1 (sl_1) seem to be more suitable. Therefore, taking DCAE−I as an example, we explored the effect of these three norms on the final anomaly detection results. The experimental results for the seven considered datasets are shown in Table III. In terms of accuracy, the l_1-norm used by the proposed method can always gain an accuracy that is prominently better than that obtained using l_2 and sl_1 for the scenes with planar targets Table III(IV–VII). And, relatively speaking, the performance of DCAE−I with the l_2-norm is better than that of DCAE−I with the sl_1-norm.

TABLE III

AUCs Obtained Using Different Types of l_{norm} in the Reconstruction Loss Term of DCAE−I for the Seven Considered Datasets. (I) HYDICE Urban. (II) Pavia Center. (III) Hyperion. (IV) Gulfport. (V) Los Angeles. (VI) San Diego I. (VII) San Diego II. Best Results in Bold and Second-Best Results Underlined

Loss Norm	I	II	III	IV	V	VI	VII
l_2	<u>0.9982</u>	0.9996	0.9983	0.9977	<u>0.9857</u>	<u>0.9904</u>	0.9919
sl_1	0.9971	<u>0.9996</u>	<u>0.9985</u>	<u>0.9978</u>	0.9826	0.9898	<u>0.9940</u>
l_1	**0.9984**	**0.9997**	**0.9992**	**0.9995**	**0.9884**	**0.9937**	**0.9945**

C. Parameter Sensitivity Analysis

In the proposed DCAE−L method, λ is used to adjust the balance of the weight-guidance term and construction residual term in the loss function. This section tests the performance of DCAE−L with various values of parameter λ on the considered datasets. As shown in Fig. 13, it is evident that the precision of DCAE−L remains consistently high as the parameter λ increases for the Pavia Center, Hyperion, and Gulfport datasets. Conversely, on the HYDICE Urban and Los Angeles datasets, DCAE−L exhibits a slight decline in precision. And it shows an initial increase in precision, followed by a subsequent decline for the two San Diego datasets. This observation indirectly indicates that the weight-guidance term incorporated into the loss function indeed contributes to anomaly detection.

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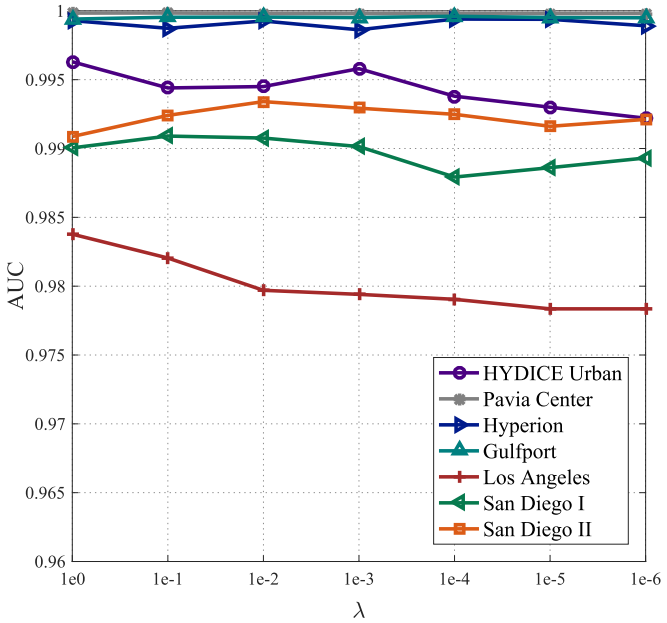


Fig. 13. AUC of the proposed DCAE−L with various values of parameter λ on the considered datasets.

VI. Conclusion

This article proposes a novel DCAE for hyperspectral anomaly detection with three important parts: the adaptive spatial feature extractor, the external feature extractor, and the background guidance module, implemented with two strategies: weight initialization (DCAE−I and added loss terms DCAE−L). Extensive experiments with seven widely used benchmark datasets have shown the superiority of the proposed methods, which can be summarized by the following aspects: 1) the spatial features' extractor introducing the DC into the regular convolution can capture elaborated spatial structure, in particular in scenes with planar targets; 2) the external feature extractor further adopts long-distance relationships with the help of NLC and a well-designed mask to reconstruct the background and suppress anomalies; 3) based on the idea of an interpretable linear reconstruction model, the designed background guidance module promotes background recovery in the decoder and makes the network more stable; and 4) moreover, we demonstrate that the l_1-norm is more effective for anomaly detection tasks. Inevitably, the proposed network also suffers from a shortcoming: it is prone to judge background classes with suspected anomalous features as anomalies, as revealed in our experiments with the Los Angeles and San Diego datasets. To address this issue, several strategies will be considered in future research, such as using a high-dimensional nonlinear space to facilitate the separation of background and anomalies, or improving the popular transformer mechanism to specifically adapt it to anomaly detection tasks.

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